PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people
AUTHORS	Ayano, Getinet; Belete, Asmare; Duko, Bereket; Tsegay, Light; Dachew, Berihun Assefa

VERSION 1 – REVIEW

REVIEWER	Dr Adam Searby Deakin University, Melbourne, Australia
REVIEW RETURNED	13-Jul-2020

GENERAL COMMENTS	Thank you for the opportunity to review this paper. Although the topic is of interest and adds to work in the field, the paper needs structural editing to reduce grammatical and spelling errors and improve readability. There are several sections in the paper that need editing work to improve clarity; this includes tables and figures that contain errors. It may be worth seeking the assistance of a professional editor to assist. Further areas needing attention: Page 2: There are several spelling mistakes in the title page. I am also curious to know why several authors are using generic email rather than their institutional email. Page 3, line 41: Is this confidence interval reported correctly? Page 5, line 12: This paragraph needs editing to improve clarity. It becomes difficult to follow your argument here; likewise, the recommended solution of a systematic review is odd given you seem to be arguing a lack of primary research. Page 5, line 47: Was this an established protocol? You should name it. Page 5, line 55: The search term paragraph would be better reported in a table or figure. Page 6, line 40: The sentence “Scopus and Web of Science were searched using terms and words suitable for the databases,” needs to be expanded for clarity. Were all the above terms used in both databases or different terms used? This needs to be made explicitly clear for a rigorous review. Page 11, line 5: You have referred to this rate being notably higher than the general population several times prior to this. Although this is likely true, you need to provide a figure and location for ‘general population’ for this assumption earlier in the paper. Page 11, line 46: This is awkwardly worded. Do you mean a quarter?

	Page 11, line 36: What about implications in the care of these people? You touch on this, but your recommendation is very broad. Page 11, line 50: Your conclusion is very brief and could do a better job of highlighting the strengths of your work. Page 21, line 3: This table needs a fair bit of work to reach a publishable standard. There are several errors in it. It is also very brief in summarising each paper; it may be worth examining some previously conducted systematic reviews to look at tables that do an excellent job of summarising the research in more detail. This is very important as this table is considered vital to many readers. Page 26, line 3: Rather than "Q1" as a header in the table, it would be better to spell out each question in the tool in each cell rather than using a key. This would improve the utility of the table for readers unfamiliar with the quality questions asked by this tool. There are also errors in this table that need correcting.

REVIEWER	Stefan Gutwinski, Co Reviewer: Karl Deutscher Charité -University, Berlin, Germany We perform a similar review with a broader perspective on prevalence of all psychiatric disorders among homeless people.
REVIEW RETURNED	21-Jul-2020

GENERAL COMMENTS	The systematic and metaanalysis reports the prevalence of depressive symptoms, mayor depression, dysthymia in homeless population. The topic is highly relevant for the majority of readers of BMJ open. The text is very well written. Statistics are well performed, but not all necessary details are reported. There are some concerns: 1) The authors missed at least some publication, which question the quality of a systematic search, for example: -Smith EM, North CS, Spitznagel EL. A systematic study of mental illness, substance abuse, and treatment in 600 homeless men. Ann Clin Psychiatry [Internet]. 1992;4(2):111–20 -Madianos MG, Chondraki P, Papadimitriou GN. Prevalence of psychiatric disorders among homeless people in Athens area: A cross-sectional study. Soc Psychiatry Psychiatr Epidemiol [Internet]. 2013 Aug;48(8):1225–34. -Sclare PD. Psychiatric disorder among the homeless in Aberdeen. Scott Med J [Internet]. 1997;42(6):173–7 2) The authors included only a female group of Strehlau V, Torchalla I, Li K, Schuetz C, Kraus M. Mental health, concurrent disorders, and health care utilization in homeless women. J Psychiatr Pract [Internet]. 2012 Sep;18(5):349–60. The same population has also a male group, published in: Krausz RM, Clarkson AF, Strehlau V, Torchalla I, Li K, Schuetz CG. Mental disorder, service use, and barriers to care among 500 homeless people in 3 different urban settings. Soc Psychiatry Psychiatr Epidemiol [Internet]. 2013 Aug;48(8):1235–43. This missed group could question the quality of the systematic search. 3) The authors do not distinguish between prevalence of life-time diagnosis of depression, dysthima etc. and prevalence of shorter time periods (e.g. 1-month prevalence). Therefore, the metaanalysis is a mix-up of both. This is very unusual. 4) Authors do not distinguish the very different subgroups of homeless population: adults, adolescence, only with trauma-experience, only with HIV, etc.

	This is of high relevance and is usually reported in similar metaanalysis. 5) The authors do not report type of statistical analysis of the subgroup and sensitivity analysis.

REVIEWER	Jean Zhuo Wang University of Ottawa, Ontario, Canada
REVIEW RETURNED	22-Jul-2020

GENERAL COMMENTS	The authors have provided an overall methodologically strong research paper addressing an important gap in research. The writing is quite clear and succinct. There are some areas that require further development as well as clarification, as follows: The authors state that they followed PRISMA guidelines with a predefined protocol - has this protocol been previously published and if yes, please list the link in this paper? Any changes to the analysis determined apriori must be transparently listed in the paper as well. The authors list the eligibility criteria for studies but did not list how they were selected. Were there two independent reviewers in study selection? If so, what was the process in resolving discrepancies amongst reviewers? There seems to be a typo on page 8 line 36 for the 95% CI - is it supposed to be 620.32? The differing definitions in of the psychiatric disorders, from study to study (ie. DSM vs MINI), require attention as it will affect the prevalence rates determined in each individual study. This raises the question of whether it is appropriate to pool these results despite these differing definitions. Is there existing literature that helps elucidate whether we expect the rates to change with the various definitions? The majority of included studies performed in high income countries was well described in the results and discussion. The authors are transparent and performed a thorough analysis of heterogeneity. It seems there is significant heterogeneity in the studies and perhaps deserves a more detailed comment regarding the effects of this on the conclusions of the paper.

REVIEWER	Cristina Maria Bostan Faculty of Psychology and Educational Sciences, „Alexandru Ioan Cuza” University, Iasi, Romania
REVIEW RETURNED	20-Aug-2020

GENERAL COMMENTS	First of all, I think this paper supports the need to have an extended view about the prevalence of depression and associated pathological problems for homeless people and the authors have made a great work in systematically reviewing the studies in the literature. Therefore, from a theoretical point of view, the review itself is done pretty accurate and detailed. The goals of the research are clear and the methods are appropiate. Still, my major concern is that there are not enough filters in the analysis of the results extracted from the studies. For example, authors should consider having the analysis done again by

	considering the type of instruments used in the studies or by the number of participants in the study. I recommend also having multiple metaanalysis, similar to their work for each pathological problems (symptoms, major depression, etc.) in order to have a more accurate image of results. Another way of analysing the data is by narrowing down even more the number of studies included here by considering the filters recommended. It is common to have multiple metaanalysis in a research, so I strongly recommend this for your research goals. Otherwise, the results are limited and could be a result of bias. These problems are also visible in the Funnel Plot of Standard Error where you can see that many of the studies included are clustered and very different. Also, authors should pay more attention to how to interpret symmetry of the funnel plots. Failure to reflect more on the type of the studies, the tools used in the studies (self-report, type of tool, interview or observation), the number of participants and also the type of studies show that the authors are not considering all of the relevant aspects of this research question. Finally, I recommend using supplemental materials for this research where authors can include for example the details about the selection of studies process or additional analysis done (agreeement of raters, etc.).

VERSION 1 – AUTHOR RESPONSE

Reviewer #1 Dr Adam Searby
Institution and Country: Deakin University, Melbourne, Australia
Thank you for the opportunity to review this paper. Although the topic is of interest and adds to work in the field, the paper needs structural editing to reduce grammatical and spelling errors and improve readability. There are several sections in the paper that need editing work to improve clarity; this includes tables and figures that contain errors. It may be worth seeking the assistance of a professional editor to assist.
Page 2: There are several spelling mistakes in the title page. I am also curious to know why several authors are using generic email rather than their institutional email.
We have addressed these issues accordingly.
Page 3, line 41: Is this confidence interval reported correctly?
No, there was an editorial issue. We revised as 26.24% (95%CI 21.02-32.22). (See page 2 abstract section line 44).
Page 5, line 12: This paragraph needs editing to improve clarity. It becomes difficult to follow your argument here; likewise, the recommended solution of a systematic review is odd given you to seem to be arguing a lack of primary research.
We have addressed these concerns accordingly. We have revised as the following “However, to the best of our knowledge, there is no previous systematic review and meta-analysis that measured the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people. This study fills this gap in the literature by analyzing the global prevalence estimates of those depressive problems among homeless people. The finding from such analysis will provide robust data on depressive problems among homeless individuals, which in turn helps policy-makers and program managers in developing policy solutions and designing appropriate and effective intervention strategies for those population groups”. (See page 4 background section paragraph 3 line 87-93).
Page 5, line 47: Was this an established protocol? You should name it.
Although the protocol was not published, we had prepared the study protocol before conducting the study. We have removed the sentence regarding protocol assuming that is may mislead some

readers.
Page 5, line 55: The search term paragraph would be better reported in a table or figure. Page 6, line 40: The sentence “Scopus and Web of Science were searched using terms and words suitable for the databases,” needs to be expanded for clarity. Were all the above terms used in both databases or different terms used? This needs to be made explicitly clear for a rigorous review.
We revised the paragraph accordingly. We also included a supplementary material for the search strategy in each database. Here is the revised statement “A systematic search of relevant studies was carried out in three reputable databases ─PubMed, Scopus, and Web of Science using keywords and combinations tailored to each database. The details about the searching technique are in supplementary material 1. In addition, the reference lists of included studies were hand-searched to identify additional articles”. (See page 4 method section line 97-101).
Page 11, line 5: You have referred to this rate being notably higher than the general population several times prior to this. Although this is likely true, you need to provide a figure and location for ‘general population’ for this assumption earlier in the paper.
We have addressed this concern accordingly. We included the following in the introduction sections “These prevalence rates are higher than the estimates in the general population (5-7.2%)”.(See page 3 introduction section paragraph 4 line 69-76).
Page 11, line 46: This is awkwardly worded. Do you mean a quarter? Page 11, line 36: What about implications in the care of these people? You touch on this, but your recommendation is very broad.
We addressed these concerns accordingly we clarified as “Finally, given the high prevalence rates of depressive problems and their potential negative consequences, early screening and interventions should be considered among those population groups. The intervention strategies might include the integration of homelessness, mental health, physical health, as well as alcohol and drug use, and other human services, working in a coordinated and joined-up way to deliver tailored and holistic interventions for those population groups” (See page 12-13 discussion section line 331-336).
Page 11, line 50: Your conclusion is very brief and could do a better job of highlighting the strengths of your work.
We have included the limitation and strength section in the revised version. See below:
Strength and limitations
Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.
Nevertheless, our study has some limitations: firstly, the vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates. Secondly, we included only peer-reviewed published articles so that book reviews, gray literature, editorials, and letters were excluded from the study during the selection process. Thirdly, studies published in the English language are only included, which suggests that relevant studies published in another language might be missed. In fact, our evaluation of possible publication bias (small study effects due to possible missingness) revealed that the funnel plot was symmetric and Egger’s regression tests provided no evidence of substantial publication bias for the prevalence of depressive symptoms, dysthymia, as well as major depressive disorders among the homeless people. Fourthly, the observed heterogeneity across the studies used to measure depressive problems was the other limitation of the current meta-analysis. To further detect the potential source of heterogeneity we conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure the outcome, and the quality of the included studies. Our analysis revealed that the quality of the studies and the sample size explained some of the observed heterogeneity ($P<0.05$). Nevertheless, the results should be interpreted with caution. Fifthly, we were unable to conduct sensitivity analysis stratified by the type of comorbid mental or substance use disorders, trauma-experiences, as well comorbid medical conditions including HIV/AIDS due to a limited number of studies in each subgroup. The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. Epidemiological evidence indicates that the involvement of more than one reviewer in the entire

process could increase the probability of including more studies in the final review [1]. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant. (See page 12 discussion section line 296-326).
Page 21, line 3: This table needs a fair bit of work to reach a publishable standard. There are several errors in it. It is also very brief in summarising each paper; it may be worth examining some previously conducted systematic reviews to look at tables that do an excellent job of summarising the research in more detail. This is very important as this table is considered vital to many readers.
Revised accordingly.
Page 26, line 3: Rather than "Q1" as a header in the table, it would be better to spell out each question in the tool in each cell rather than using a key. This would improve the utility of the table for readers unfamiliar with the quality questions asked by this tool. There are also errors in this table that need correcting.
We edited the errors accordingly. Regarding the abbreviation-related concerns, which is a very acceptable suggestion but the questions are so long and it's difficult to include them in the table. We also tried to read prior studies that used the same tool and we found that roughly all studies used abbreviating for that.

Reviewer #2
Reviewer Name: Stefan Gutwinski, Co Reviewer: Karl Deutscher
Institution and Country: Charité -University, Berlin, Germany
Please state any competing interests or state 'None declared': We perform a similar review with a broader perspective on prevalence of all psychiatric disorders among homeless people.

The systematic and metaanalysis reports the prevalence of depressive symptoms, mayor depression, dysthymia in homeless population. The topic is highly relevant for the majority of readers of BMJ open. The text is very well written. Statistics are well performed, but not all necessary details are reported.
There are some concerns:
1) The authors missed at least some publication, which question the quality of a systematic search, for example:
-Smith EM, North CS, Spitznagel EL. A systematic study of mental illness, substance abuse, and treatment in 600 homeless men. Ann Clin Psychiatry [Internet]. 1992;4(2):111–20
-Madianos MG, Chondraki P, Papadimitriou GN. Prevalence of psychiatric disorders among homeless people in Athens area: A cross-sectional study. Soc Psychiatry Psychiatr Epidemiol [Internet]. 2013 Aug;48(8):1225–34.
-Sclare PD. Psychiatric disorder among the homeless in Aberdeen. Scott Med J [Internet]. 1997;42(6):173–7
Thank you very much for the important suggestions and for providing us those relevant studies. However, previously, we did not include those studies in the final analysis because of the following reasons:
(1) the third study (-Sclare PD. Psychiatric disorder among the homeless in Aberdeen. Scott Med J [Internet]. 1997;42(6):173–7) has not measured the prevalence rates of depressive disorders or dysthymia in the result. The reported other mental disorders including schizophrenia, bipolar disorders, substance use disorders, and so on.
(2) The second study (Madianos MG, Chondraki P, Papadimitriou GN. Prevalence of psychiatric disorders among homeless people in Athens area: A cross-sectional study. Soc Psychiatry Psychiatr Epidemiol [Internet]. 2013 Aug;48(8):1225–34) has reported the current prevalence of major depressive disorders (2 weeks) and the 1 year or 2-year prevalence of dysthymia. Nevertheless, our

study focuses on the lifetime prevalence of those disorders;
(3) Regarding the first study we found only the abstract and the full text is not accessible during our search may be due to the year of publication very long (1992) and the journal has stopped publishing studies several years ago. We tried to reach out to the authors but they were not responding.
In addition, as you already know missing some studies could be expected in systematic review and meta-analysis and we conducted publication bias analysis to evaluate the effect of missed studies (if any) in our results. Notably, the analysis revealed that the effect of the missed study (if any) was not statistically significant (there is no evidence of significant publication bias)
2) The authors included only a female group of Strehlau V, Torchalla I, Li K, Schuetz C, Kraus M. Mental health, concurrent disorders, and health care utilization in homeless women. J Psychiatr Pract [Internet]. 2012 Sep;18(5):349–60.
The same population has also a male group, published in Krausz RM, Clarkson AF, Strehlau V, Torchalla I, Li K, Schuetz CG. Mental disorder, service use, and barriers to care among 500 homeless people in 3 different urban settings. Soc Psychiatry Psychiatr Epidemiol [Internet]. 2013 Aug;48(8):1235–43.
This missed group could question the quality of the systematic search.
Thank you very much for the extensive review and for providing us the missed study. We now included that study in the analysis. The findings were not significantly changed after including the suggested study, which is consistent with our publication bias analysis report that suggested the effect of missed study (if any) was not statistically significant.
Here is the revised prevalence estimate after including the above study:
Major depressive disorders:
The pooled prevalence of major depressive disorders was 26.24% (95%CI 21.02-32.22). Significant heterogeneity has been found for this analysis (I^2=97.57; $p < 0.001$) (figure 3). (See page 8 result section line 197-199).
Persistent depressive disorders (dysthymia)
Our random-effects meta-analysis resulted in pooled prevalence estimates of dysthymia in the homeless people 8.25% (95% CI; 4.79-11.86). This analysis also resulted in a significant heterogeneity between the studies (I^2=89.80%; $p < 0.001$) (figure 4). (See page 8 result section line 201-203).
Regarding the inclusion of the above study, which was conducted involving female participants only, we included it in the final analysis because as you already know in a meta-analysis (random effect meta-analysis) the samples of both male and female participants across all studies have been pooled together to a single estimate (average of all male and female participants). In addition, based on your suggestions now we have repeated the analysis removing the study which was conducted by involving only female participates (n=1) from the model, and the result was roughly the same as the main analysis (26.24%vs. 26.13%). This result suggests the robustness of the finding of the current study.
We also included the following statement in the result section “Finally, one of the studies included in the final analysis was conducted among female participants only. When we remove this study from the final analysis, the prevalence estimates remained virtually the same 26.13% (95%CI 20.76-32.33)”. (See page 9 result section line 234-236).

3) The authors do not distinguish between the prevalence of life-time diagnosis of depression, dysthima etc. and prevalence of shorter time periods (e.g. 1-month prevalence). Therefore, the metaanalysis is a mix-up of both. This is very unusual.
The prevalence of depressive disorders and dysthymia represents the lifetime prevalence whereas the prevalence of depressive symptoms was current prevalence. We have included the following statement in the method section to clarify this:
Definition of key terms and concepts
Depressive disorders and dysthymia
This systematic review and meta-analysis combined studies from a wide range of settings and

countries across the globe that reported the lifetime prevalence estimate of major depressive disorders as well as dysthymia according to standard diagnostic criteria's, which were assessed by using standard diagnostic instruments such as the Diagnostic Statistical Manual of Mental disorders (DSM), Schedule for Clinical Assessment of Neuropsychiatry (SCAN), International Classification of Disease (ICD), and Composite International Neuropsychiatric Interview (CIDI) and Mini-International Neuropsychiatric Interview (MINI). (See page 5 method section line 115-123).
Depressive symptoms
The prevalence of depressive symptoms in this review indicates the current prevalence as measured by the screening instruments such as the Centre for Epidemiologic Surveys for Depression (CES-D), Depression, Anxiety, and Stress Scale (DASS), Patient Health Questionnaire (PHQ-9), and Zung Self-rating Depression Scale. (See page 5 method section line 124-128).
4) Authors do not distinguish the very different subgroups of homeless population: adults, adolescence, only with trauma-experience, only with HIV, etc.
This is of high relevance and is usually reported in similar metaanalysis.
We did not conduct a subgroup analysis for those population groups specifically due to a small number of studies in each subgroup. we have included the following in the implication sections future studies "Fifthly, we were unable to conduct sensitivity analysis stratified by the type of comorbid mental or substance use disorders, trauma-experiences, as well comorbid medical conditions including HIV/AIDS due to a limited number of studies in each subgroup". (See page 12 discussion section line 316-318).
However, we have conducted a subgroup analysis based on the age of the participants, the quality of the studies, the instrument used to measure depressive problems, as well as the sample size. This analysis suggested that the prevalence estimates of both depressive symptoms and depressive disorders were found to higher in studies with lower quality, small sample, and older participants when compared with their counterparts. (See page 8-9 result section line 204-236).
5) The authors do not report type of statistical analysis of the subgroup and sensitivity analysis.
We now included the statistic for all subgroups. All the subgroup analyses were conducted using a random-effect model. (See page 8 result section line 208-209).

Reviewer:3 3
Reviewer Name: Jean Zhuo Wang
Institution and Country: University of Ottawa, Ontario, Canada
Please state any competing interests or state 'None declared': None declared.
The authors have provided an overall methodologically strong research paper addressing an important gap in research. The writing is quite clear and succinct. There are some areas that require further development as well as clarification, as follows:
Q1. The authors state that they followed PRISMA guidelines with a predefined protocol - has this protocol been previously published and if yes, please list the link in this paper? Any changes to the analysis determined apriori must be transparently listed in the paper as well.
Although the protocol was not published, we had prepared the study protocol before conducting the study. We have removed the sentence regarding protocol assuming that is may mislead some readers.
Q2. The authors list the eligibility criteria for studies but did not list how they were selected. Were there two independent reviewers in study selection? If so, what was the process in resolving discrepancies amongst reviewers?
We have addressed this concern accordingly; we have included the following in the method and discussion sections:
Selection of studies for the inclusion in the systematic review and meta-analysis
In this systematic review and meta-analysis, the corresponding author (GA) identified articles and consequently evaluated them by their titles and abstract based on the eligibility criteria. Full-text articles were selected for further assessment by the author. This author further appraised the full text

of each study and subsequently retained those full-text articles to be included in the final analysis. (See page 5 method section line 109-114).
We also included the following in the discussion section: The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. Epidemiological evidence indicates that the involvement of more than one reviewer in the entire process could increase the probability of including more studies in the final review [1]. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant. (See page 12 result discussion line 319-326).
In fact, quality assessment and data extraction were conducted independently by two reviewers, which is the common practice in the vast majority of published systematic reviews.
Q3. There seems to be a typo on page 8 line 36 for the 95% CI - is it supposed to be 620.32?
Edited accordingly.
q4. The differing definitions in of the psychiatric disorders, from study to study (ie. DSM vs MINI), require attention as it will affect the prevalence rates determined in each individual study. This raises the question of whether it is appropriate to pool these results despite these differing definitions. Is there existing literature that helps elucidate whether we expect the rates to change with the various definitions?
We have addressed this concern accordingly. We have conducted a sensitivity analysis based on the instrument used to measure depressive disorders and the results revealed no statistically significant variation in the prevalence across the tools. We also included the following in the result section “We also found that the prevalence of depressive disorders was highest when measured by MINI (33.64%) followed by CIDI (27.42%), SCID (21.66%), and DSM (20.18%). However, the observed differences across the tools were not statistically significant (P>0.005)”. (See page 9 result section line 227-229).
q5. The majority of included studies performed in high income countries was well described in the results and discussion. The authors are transparent and performed a thorough analysis of heterogeneity. It seems there is significant heterogeneity in the studies and perhaps deserves a more detailed comment regarding the effects of this on the conclusions of the paper.
We have addressed the concerns. We included the following in the discussion section “Fourthly, the observed heterogeneity across the studies used to measure depressive problems was the other limitation of the current meta-analysis. To further detect the potential source of heterogeneity we conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure the outcome, and the quality of the included studies. Our analysis revealed that the quality of the studies and the sample size explained some of the observed heterogeneity (P<0.05). Nevertheless, the results should be interpreted with caution”. (See page 912 discussion section line 310-315).

Reviewer: 4
Reviewer Name: Cristina Maria Bostan, Institution and Country: Faculty of Psychology and Educational Sciences, „Alexandru Ioan Cuza” University, Iasi, Romania
Please state any competing interests or state ‘None declared’: None
First of all, I think this paper supports the need to have an extended view about the prevalence of depression and associated pathological problems for homeless people and the authors have made a great work in systematically reviewing the studies in the literature. Therefore, from a theoretical point of view, the review itself is done pretty accurate and detailed. The goals of the research are clear and the methods are appropiate.
Still, my major concern is that there are not enough filters in the analysis of the results extracted from the studies. For example, authors should consider having the analysis done again by considering the type of instruments used in the studies or by the number of participants in the study. I recommend also having multiple metaanalysis, similar to their work for each pathological problems (symptoms,

major depression, etc.) in order to have a more accurate image of results. Another way of analysing the data is by narrowing down even more the number of studies included here by considering the filters recommended. It is common to have multiple metaanalysis in a research, so I strongly recommend this for your research goals. Otherwise, the results are limited and could be a result of bias. These problems are also visible in the Funnel Plot of Standard Error where you can see that many of the studies included are clustered and very different. Also, authors should pay more attention to how to interpret symmetry of the funnel plots. Failure to reflect more on the type of the studies, the tools used in the studies (self-report, type of tool, interview or observation), the number of participants and also the type of studies show that the authors are not considering all of the relevant aspects of this research question.
Finally, I recommend using supplemental materials for this research where authors can include for example the details about the selection of studies process or additional analysis done (agreeement of raters, etc.).
We thank you so much for the reviewers for the kind words. We have addressed the concerns accordingly. We have included additional sensitivity analysis based on the tools used and the sample sized. We also included the following in the result section: (See page 9 result section line 227-236).
We also found that the prevalence of depressive disorders was highest when measured by MINI (33.64%) followed by CIDI (27.42%), SCID (21.66%), and DSM (20.18%). However, the observed differences across the tools were not statistically significant (P>0.05).
Moreover, our sensitivity analysis based on the sample size revealed that the prevalence estimates of both depressive symptoms as well as depressive disorders were found to considerably higher in studies that involved a lower small sample size when compared with studies that involved a higher sample size.
Finally, one of the studies included in the final analysis was conducted among female participants only. When we remove this study from the final analysis, the prevalence estimates remained virtually the same 26.13% (95%CI 20.76-32.33).
Regarding the selection process, we have included the following in the method and discussion sections:
Selection of studies for the inclusion in the systematic review and meta-analysis
In this systematic review and meta-analysis, the corresponding author (GA) identified articles and consequently evaluated them by their titles and abstract based on the eligibility criteria. Full-text articles were selected for further assessment by the author. This author further appraised the full text of each study and subsequently retained those full-text articles to be included in the final analysis. (See page 5 method section line 109-114).
We also included the following in the discussion section: The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. Epidemiological evidence indicates that the involvement of more than one reviewer in the entire process could increase the probability of including more studies in the final review [1]. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant. (See page 12 result discussion line 319-326).
In fact, quality assessment and data extraction were conducted independently by two reviewers, which is the common practice in the vast majority of published systematic reviews.

VERSION 2 – REVIEW

REVIEWER	Dr Adam Searby Institute for Health Transformation, Deakin University, Geelong, Australia
REVIEW RETURNED	21-Oct-2020

GENERAL COMMENTS	Thank you for attending to revisions as recommended by the reviewers - they have improved the paper's readability. There are still some minor grammatical errors that could be addressed in the typesetting stage prior to publication. Can you please clarify if only one author conducted the selection and screening of papers for the systematic review? It reads as if this is the case (page 5, line 109), and if so, there needs to be more justification of the rationale for this approach as it can be considered unconventional. I can see that you have addressed this point later in the paper, however If this point can be clarified I believe the paper is suitable for publication.

REVIEWER	Jean Zhuo Wang Faculty of Medicine, University of Ottawa, Ottawa, ON, Canada
REVIEW RETURNED	25-Oct-2020

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript again. The authors have done a good job of addressing previous comments. Thank you for addressing the comments regarding article selection. This is a concern that there is only one reviewer however this has been described in the discussion satisfactorily. The additional sensitivity analyses are a strength to the paper. Is it a typo that you listed p>0.005 being not statistically significant or is it truly around 0.005? (page 9, line 229) Your p value for statistical significance was set at 0.05. Please add the study protocol sentence again and mention that it was not published apriori or add it to the supplemental materials. It is important for readers to know whether you've had a protocol, and whether you have changed anything in the methodology after the creation of the protocol to identify areas of bias. If there were any key changes in the methodology from the protocol to the final paper, please describe them in the paper. Some new comments have arisen with changes and comments from other reviewers. There are still concerns regarding the writing quality of the paper. In table 1, page 22, what does it mean by "adults with mental lines - topolovec-vranic", "adults VETRIAN". Also, consider using a less stigmatizing term from "mentally ill" such as adults with mental illness. Also please provide more consistent use of caps - for instance sometimes countries are listed as "Canada" and sometimes are listed as "CANADA". For Table 1, I would also encourage more details to be provided regarding the study population. For instance, are these adults experiencing homelessness or just any adults? Many of these studies utilize different definitions for "homelessness" as well, and I would recommend providing slightly more details, ie. for Okamura "adults experiencing homelessness as defined by the McKinney-Vento Act". It is helpful to have the tools used, but also would be helpful to list which depressive disorder was measured "ie. dysthymia, MDD" - that will help readers better interpret the % listed in the outcome column. What is meant by the term "depressive problems"? This is not a well defined term and would prefer the use of "depressive disorders" throughout the manuscript.

REVIEWER	Cristina Maria Bostan Faculty of Psychology and Educational Sciences, „Alexandru Ioan Cuza" University, Iasi (ROMANIA)
REVIEW RETURNED	02-Nov-2020

GENERAL COMMENTS	Comments I believe most of the comments were analyzed and the authors made a great effort in responding to each of my comments. Although the manuscript has improved considerably, here are some recommendations to follow in drafting a final version: 1. use simple and plain language expressions to highlight results. Although they are of great relevance, sometimes authors tend to highlight them by using expression that could distort the scientific nature of the results. For example, at page 3, in the Conclusions area authors use the expression „remarkably higher than the reported prevalence rates in the general population". This word „remarkably„ has no place there because it has no scientific meaning (intensity of coefficients, the magnitude of coefficients, percentage, etc.).

VERSION 2 – AUTHOR RESPONSE

Reviewer #1 Dr Adam Searby
Institution and Country: Deakin University, Melbourne, Australia
Thank you for attending to revisions as recommended by the reviewers - they have improved the paper's readability. There are still some minor grammatical errors that could be addressed in the typesetting stage prior to publication.

Q1. Can you please clarify if only one author conducted the selection and screening of papers for the systematic review? It reads as if this is the case (page 5, line 109), and if so, there needs to be more justification of the rationale for this approach as it can be considered unconventional. I can see that you have addressed this point later in the paper, however If this point can be clarified I believe the paper is suitable for publication.
Author reply: Thank you. We have clarified this issue as follows: "The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant. In addition, we have involved two independent investigators for data extraction and quality assessment. The existing guidelines in systematic review and meta-analysis including Cochrane library and PRISMA suggest quality assessment and data extraction must be performed by two independent investigators [26, 81]. However, involving the second author in entire processes including selection and screening is not mandatory (optional) [26, 81]. In fact, a recent systematic review indicated that the involvement of more than one reviewer in the entire process could increase the probability of including more studies in the final review [82]. Another systematic review on the subject revealed no significant differences in the rates of effectiveness (avoiding the number of studies with inappropriate exclusions) in studies that involved a second author in the entire process and studies that did not [83]". ". (See page 12-13 strength and limitation section of the discussion paragraph 3 line 328-342).

Reviewer#3: Jean Zhuo Wang
Institution and Country: University of Ottawa, Ontario, Canada
Thank you for the opportunity to review this manuscript again.

The authors have done a good job of addressing previous comments. Thank you for addressing the comments regarding article selection. This is a concern that there is only one reviewer however this has been described in the discussion satisfactorily.
Q1. The additional sensitivity analyses are a strength to the paper. Is it a typo that you listed p>0.005 being not statistically significant or is it truly around 0.005? (page 9, line 229) Your p value for statistical significance was set at 0.05.
Author reply: Thank you. Yes, it is a typo error. It should be P>0.05.
Q1. Please add the study protocol sentence again and mention that it was not published apriori or add it to the supplemental materials. It is important for readers to know whether you've had a protocol, and whether you have changed anything in the methodology after the creation of the protocol to identify areas of bias. If there were any key changes in the methodology from the protocol to the final paper, please describe them in the paper.
Author reply: we have addressed this comment accordingly. We included the following sentence in the method section for clarification "We used a predesigned protocol (unpublished) for searching, extracting data, inclusion or exclusion of studies, assessing quality, and data analysis." (See page 4 method section line 97-99).

Q2. Some new comments have arisen with changes and comments from other reviewers. There are still concerns regarding the writing quality of the paper. In table 1, page 22, what does it mean by "adults with mental lines - topolovec-vranic", "adults VETRIAN". Also, consider using a less stigmatizing term from "mentally ill" such as adults with mental illness. Also please provide more consistent use of caps - for instance sometimes countries are listed as "Canada" and sometimes are listed as "CANADA".
Author reply: We have fixed the suggested editorial issues accordingly. (See Table 1 and Table 2 page 23-28 line 611-619)
Q3. For Table 1, I would also encourage more details to be provided regarding the study population. For instance, are these adults experiencing homelessness or just any adults? Many of these studies utilize different definitions for "homelessness" as well, and I would recommend providing slightly more details, ie. for Okamura "adults experiencing homelessness as defined by the McKinney-Vento Act". It is helpful to have the tools used, but also would be helpful to list which depressive disorder was measured "ie. dysthymia, MDD" - that will help readers better interpret the % listed in the outcome column.
Author reply: Thank you. We have included the definition of homeless in the method section (See page 4-5 method section 1 line 131-137) and the types of depressive problems in Table1 (See Table 1 page 23-26 line 611-615))
Here is the definition of homelessness:
Homelessness
There is no internationally consistent definition of homelessness. Some scholars (countries) define homelessness narrowly as the absence of access to safe shelter while others define broadly comprising those people living in marginal housing (marginal accommodations), people who are sleeping in public places or designed shelter, and rooflessness [27, 28]. In the present study, we have included studies that defined homelessness in various ways. Thus, homelessness represents the comprehensive definition mentioned above (which is beyond rooflessness).
Q4. What is meant by the term "depressive problems"? This is not a well defined term and would prefer the use of "depressive disorders" throughout the manuscript.
Author reply: This review included both depressive disorders (MDD and dysthymia) and depressive symptoms. So, we used the term depressive problems to capture both the disorders and symptoms.

Reviewer #4: Cristina Maria Bostan
Faculty of Psychology and Educational Sciences,"Alexandru Ioan Cuza" University, Iasi (ROMANIA)
I believe most of the comments were analyzed and the authors made a great effort in responding to

each of my comments.
Although the manuscript has improved considerably, here are some recommendations to follow in drafting a final version:
1. use simple and plain language expressions to highlight results. Although they are of great relevance, sometimes authors tend to highlight them by using expression that could distort the scientific nature of the results. For example, at page 3, in the Conclusions area authors use the expression „remarkably higher than the reported prevalence rates in the general population".
This word „remarkably„ has no place there because it has no scientific meaning (intensity of coefficients, the magnitude of coefficients, percentage, etc.).
Author reply: Thank you. We have now fixed unnecessary expressions and removed the word remarkably from the entire document.

VERSION 3 – REVIEW

REVIEWER	Dr Adam Searby Deakin University Institute of Health Transformation, Melbourne, Australia
REVIEW RETURNED	24-Dec-2020

GENERAL COMMENTS	Thank you for the opportunity to review this revision, and thank you for your comprehensive inclusion of a justification of the review method used. I believe this could be a major criticism of the paper, however including this information helps to alleviate this issue. There are still some minor grammatical errors in the paper, however these would be best handled by the publisher given regional variations in style and spelling. Provided recommendations of other reviewers are met, I would recommend the paper be accepted.